

RPC (Remote Procedure Call)

1. What is RPC?

RPC (Remote Procedure Call) is a communication technique where one program calls a function/method running on another machine as if it were a local function.

You call:

```
None  
userService.getUser(101)
```

But execution happens on another server.

2. Core Idea

RPC hides network complexity.

Instead of manually handling:

```
None  
HTTP requests  
Sockets  
Serialization  
Retries  
Response parsing
```

Developer simply calls methods.

3. Example

Client code:

```
None  
paymentService.pay(orderId, amount)
```

Internally:

```
None  
Serialize request  
  
Send over network  
  
Remote server executes pay()  
  
Return response  
  
Deserialize response
```

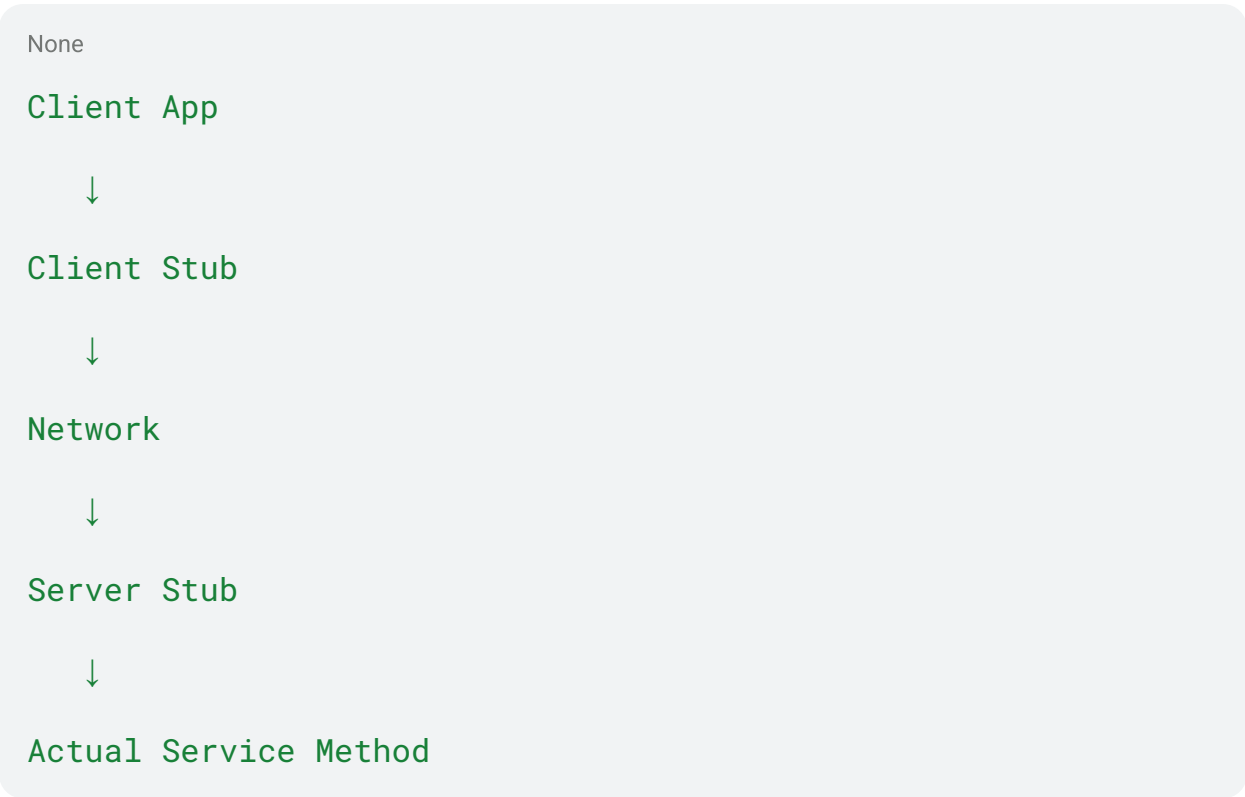
Looks like local function call.

4. Popular RPC Frameworks

Framework	Company
gRPC	Google

Thrift	Meta
RPyC	Python RPC framework

5. How RPC Works



6. Components

Component	Meaning
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Client Stub	Local proxy function
Server Stub	Receives request
IDL	Interface definition language
Serializer	Converts objects to bytes
Transport	HTTP/2, TCP etc.

7. Example Flow

None

Client calls getProduct(10)

↓

Request serialized

↓

Sent to Product Service

↓

Service fetches DB data

↓

Returns response

↓

Client gets object

8. Why RPC is Popular

None

Fast communication

Compact payloads

Strong contracts

Code generation

Low latency

Good for microservices

9. RPC vs REST Basic Difference

REST:

None

GET /users/10

POST /orders

RPC:

None

userService.getUser(10)

orderService.createOrder(...)

REST uses resources. RPC uses methods/functions.

10. Serialization Formats

Common encodings:

None

JSON

XML

CSV

Protocol Buffers

Thrift

Avro

Most modern RPC systems use binary formats.

11. Why Binary Format Helps

Compared to JSON:

None

Smaller payload size

Faster parsing

Lower CPU usage

Lower bandwidth

Very useful at scale.

12. gRPC Example

Service definition:

```
None
service UserService {
    rpc GetUser(UserRequest) returns (UserResponse);
}
```

Call:

```
None
GetUser(id=101)
```

13. Strong Schema Contracts

RPC uses schema files.

Benefits:

```
None
Auto-generate client SDK
Auto-generate server code
Clear request/response types
Version compatibility rules
```


14. Best Use Cases

Use RPC for:

None

Microservice internal communication

Real-time backend systems

Low-latency systems

Mobile backend aggregation

IoT devices

High-throughput APIs

15. Example in Microservices

None

Order Service calls Payment Service

Payment Service calls Notification Service

Inventory Service calls Warehouse Service

RPC works very well internally.

16. Why Better than REST Internally

None

Faster than JSON APIs

Typed interfaces

Smaller messages

Streaming support

Generated clients

17. Streaming Support (gRPC)

RPC frameworks may support:

None

Unary request-response

Server streaming

Client streaming

Bidirectional streaming

Useful for chat/live systems.

18. Disadvantages

None

Harder debugging than REST

Binary traffic less human-readable

Schema updates required

External third-party clients harder to integrate

Browser support weaker than REST

19. RPC vs REST vs GraphQL

Feature	RPC	REST	GraphQL
Style	Method calls	Resource URLs	Query language
Speed	High	Medium	Medium
Human readable	Low	High	High
Strong typing	High	Medium	High

Browser friendly	Medium	High	High
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20. Internal vs External APIs

Common pattern:

None

External public API → REST

Internal service communication → RPC

Very common in large companies.

21. Example HLD Use

Food delivery app:

None

Frontend → REST API Gateway

Gateway → User Service (RPC)

Gateway → Payment Service (RPC)

Gateway → Tracking Service (RPC)

22. Interview Talking Points

Say RPC when system needs:

None

High performance

Low latency

Strong contracts

Service-to-service communication

Streaming

Large scale backend traffic

23. When Not Ideal

Avoid RPC for:

None

Public APIs for many external developers

Simple CRUD internet APIs

When curl/Postman simplicity matters

REST easier there.

24. Real Example

Video platform:

None

Upload Service calls Transcoder Service

Recommendation Service calls User Profile Service

Analytics Service streams metrics

25. One-Line Summary

RPC is a communication model where applications call remote functions on other servers like local methods, enabling fast, typed, efficient service-to-service communication.